GridIQ

Feasibility data handover

Demo Corporation · Durban Distribution Ctr

Assessment
GQ-A-0004 · 7-day assessment

Report
revision 2 · issued

Observation period (SAST)
2026-11-02 21:00 -> 2026-11-09 21:00

Observed
7.00 days of 7

Site address
88 Logistics Blvd, Durban

Measurement points
PV (pv), Warehouse Main (incomer)

This revision replaces revision 1: layout fix: blank footer pages removed from the PDF

1. What is supplied

Measured data for an independent feasibility study. GridIQ does not size PV, batteries or generators and makes no savings claim; the receiving engineer applies their own tariffs, assumptions and models.

- 30-minute interval load profile (kW, kVA)
- daily energy
- billed maximum demand with timestamps
- power-quality screening results
- metering topology and observation limits

Files in the data package

File	Content
manifest.json	IDs, period, conventions, units, file list with SHA-256 checksums
load_profile_30min.csv	30-minute kW, kVA, coverage per point (period-ending rows)
readings_1min.csv	1-minute averages of the core channels per point
daily_energy.csv	SAST daily energy and peaks per point
findings.csv	rule findings with evidence and review state
*.pdf	overview and power-quality reports (where issued)

2. Metering topology and limits

Point	Role	Parent	Coverage	Import (kWh)	Max demand (kVA)
PV	pv	sub-meter	10.9 %	4 073,5	177,2
Warehouse Main	incomer	top level	100.0 %	29 455,6	403,6

sum of 1 top-level incomer(s); sub-meters are inside these and are not added

- Observation period 7 days (2026-11-02T19:00:36.240Z to 2026-11-09T19:00:36.240Z); results describe this period only. No annual figure is extrapolated.
- One-second RMS trends cannot prove that no sub-second disturbance occurred; events listed are inferred from sampled trends unless marked device-recorded.
- A single net grid meter cannot separate PV production, battery behaviour or individual equipment loads.
- PV: 11 % of expected measurements received (3 gap(s) of 30 min or more).
- PV: the meter's capability profile is not confirmed; only channels present in the data are reported.
- Warehouse Main: the meter's capability profile is not confirmed; only channels present in the data are reported.

3. Relevant power-quality concerns

- Warehouse Main: Current unbalance reached 30.41 % (95th percentile, unbalance as computed by the meter (per its configuration); not stated to be negative-sequence unbalance). Rebalancing single-phase loads usually helps.

4. Operating context and missing inputs

Still missing: declared operating hours (default 06:00–18:00 weekdays used); supply details and tariff; existing PV / generator / battery details; fault level at the point of supply; a full month or more of data (seasonal and monthly patterns).

Technical appendix

Dataset schema
gridiq.assessment-dataset/1

Dataset SHA-256
79b49ad7529769dca21be0162d67ec10…

Calculation version
gridiq-calc-2026.11.1

Rule version
gridiq-rules-2026.11.1

Template version
gridiq-templates-2026.11.1

Data cutoff (SAST)
2026-11-09 21:00

Generated (SAST)
2026-11-11 21:13

Time convention
[start, end): a value at bucket B covers B <= t < B + interval

Method

- Readings are attributed to a measurement point by deployment window at the measurement time; buckets that straddle an install or removal are recomputed from the samples inside the window.
- Energy from cumulative kWh register differences per deployment; a negative step is a register reset and never summed through; power integration only where registers are absent.
- Billed demand: 30-minute minute-weighted kVA averages; instantaneous peaks are reported separately.
- Power quality: 10-minute values screened against NRS 048-2 (screening thresholds; formal compliance needs a Class A instrument, one full week and >= 95 % valid 10-minute values).
- Sign convention: import positive; the deployment sign correction is applied. Times in the data are UTC; this report shows SAST (UTC+2).

Full SHA-256 of the dataset: 79b49ad7529769dca21be0162d67ec106d55c4fa8e310cfa3036c896474837e7